

A Software-Based Comparative Study of Error Sensitivity in Classical and Quantum Optimization Algorithms

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1 Introduction

Everyone is familiar with the idea of a classical computer, but not everyone is familiar with the idea of a quantum computer. The Classical computing processes information using bits that take definite values of 0 or 1, whereas quantum computing uses qubits that can exist in superpositions of states and become entangled, enabling fundamentally different computational behavior [?]. Quantum computing has the potential to improve the performance of certain computational tasks, particularly search and optimization problems. Many quantum algorithms promise theoretical speedups over classical methods; however, these results often rely on ideal assumptions that ignore noise, limited circuit depth, and other practical constraints of current quantum technology. As a result, there is a gap between theoretical expectations and the real-world behavior of quantum algorithms. This work aims to bridge that gap through a software-based comparison of classical and quantum optimization algorithms, focusing on their sensitivity to noise and errors using classical simulation.

This research evaluates quantum optimization algorithms under realistic execution conditions and compares them with classical heuristics. Rather than focusing on theoretical speedups, it examines practical performance, robustness, and solution quality in noisy environments. Classical methods—genetic algorithms and greedy search—are compared with quantum approaches including Grover-based search and QAOA. A modular benchmarking framework with configurable noise models enables fair, reproducible evaluation using metrics such as solution quality, stability, circuit depth, and noise sensitivity. Results show quantum performance often degrades under realistic constraints, while classical heuristics remain more robust, highlighting the importance of hybrid strategies and practical evaluation methods.

This paper is organized into five benchmark comparisons used to evaluate classical and quantum optimization algorithms across representative problem types. Each row pairs a classical algorithm with a quantum counterpart and identifies the problem instance used to compare them. The first and primary comparison evaluates a Genetic Algorithm against QAOA on the Max-Cut problem, a standard benchmark for quantum optimization that allows a fair global search comparison. The second compares a Greedy Algorithm with QAOA on the Knapsack problem, highlighting the difference between a simple heuristic and a

variational quantum method for constrained optimization. The third experiment contrasts a Genetic Algorithm with Grover’s Algorithm using MAX-SAT formulated as a search problem, emphasizing heuristic versus quadratic-speedup search. The fourth compares Greedy search with Grover’s Algorithm on Number Partitioning, illustrating structured versus unstructured search. Finally, a combined Genetic+Greedy versus QAOA+Grover evaluation on Random QUBO provides a unified benchmark across all algorithms.

2 Related Works

Even though quantum computing is relatively new, there has been a lot of work done in the field. The most relevant work to this project I found was the paper by Airton Arruda, et al. titled ”A Software-Based Comparative Study of Error Sensitivity in Classical and Quantum Optimization Algorithms” [?]. This study compares three paradigms for solving the Weighted Max-Cut problem: a Genetic Algorithm, a Graph Neural Network, and a tensor network approach using Density Matrix Renormalization Group. Evaluating graphs from 10 to 250 nodes, it analyzes solution quality, runtime, and memory usage. Results show trade-offs in scalability, efficiency, and approximation performance. Additionally, papers such as ”Introduction to Classical and Quantum Computing” by Thomas G. Wong [?] and ”Quantum Approximate Optimization Algorithm” by Leo Zhou, et al. [?] provide foundational insights into the principles and fundamentals to understand the algorithms used in this project.

3 Methodology

Dijkstra’s algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [?].

4 Results

An exemplary figure is shown in Figure ?? . Any time a figure is used, it must have a caption and be directly referred to and discussed in the text.

5 Conclusion

References

- [1] AITOR MORAIS, ENEKO OSABA, I. P. I. O. Comparative analysis of classical and quantum-inspired solvers: A preliminary study on the weighted max-cut problem.
- [2] WONG, T. G. *Introduction to Classical and Quantum Computing*. Self-published, 2019.
- [3] ZHOU, L., WANG, S.-T., CHOI, S., PICHLER, H., AND LUKIN, M. D. Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices.

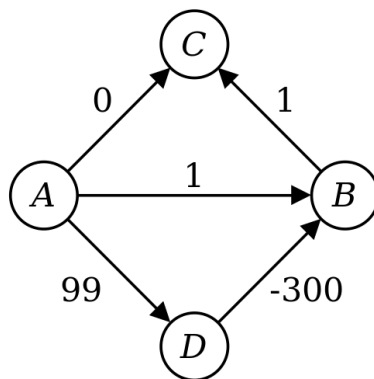


Figure 1: Dijkstra's algorithm does not work on graphs with negative edge weights, as evidenced here.